Hide and Seek: Tracing the PRDM9 Locus in Crocodilians Through Synteny

Zainab Siddiqui, Matthew Fujita, Zoe Muller, Joshua Rivera
Division of Data Science, College of Science,
The University of Texas at Arlington

THE UNIVERSITY OF TEXAS AT ARLINGTON

NCBI

Background

- What is PRDM9?
  - A gene that dictates meiotic recombination sites across many vertebrate taxons (especially mammals, snakes, and fish)
  - Can give important information on how genetic diversity occurs in a species
- Why crocs?
  - Understudied group; PRDM9 is assumed absent throughout Archosauria (i.e., crocodilians, birds, dinosaurs)
  - Few studies focus on crocodilians despite them being a relatively stable clade (evolutionarily)
- Bioinformatics/ computational approach
  - Combining biology & data science methods



Methods

- Bioinformatic techniques used:
  - Outgroup analysis with *Chrysemys picta bellii* (the Western Painted Turtle), *known* to have PRDM9, used as reference
  - **Synteny**: comparing gene order between species
- Computational tools used:
  - Python w/shell scripting & CLI
  - BLAST (Basic Local Alignment Search Tool)
  - Tiberius – deep learning model for gene annotation
- Species:
  - American alligator
  - Chinese alligator
  - Australian saltwater crocodile
  - Indian gharial
- Approx. ~10GB of genomic/FASTA data

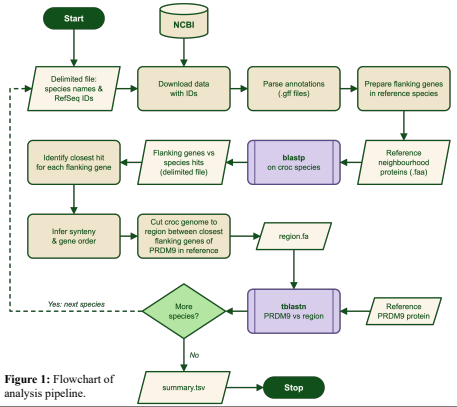


Figure 1: Flowchart of analysis pipeline.

Results

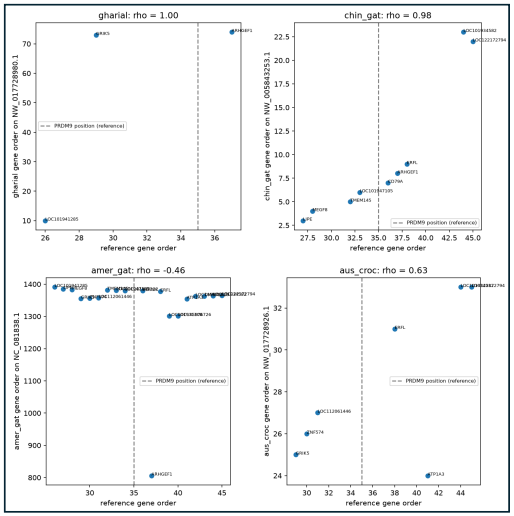


Figure 2: Synteny dot-plots on various crocodilians vs. *C. picta bellii* with labelled identified flanking genes and rho values showing genetic order similarity.

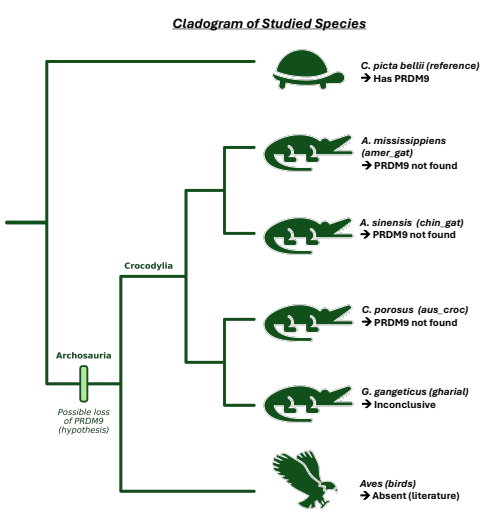


Figure 3: Cladogram depicting relationship between Archosaurs and turtle outgroup, as well as hypothesis for ancestral loss of PRDM9.

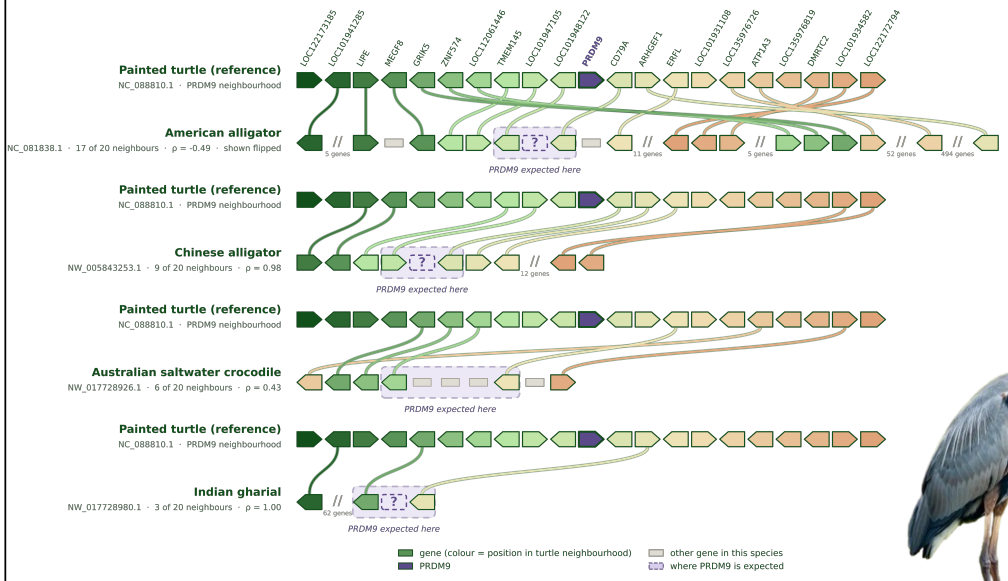


Figure 4: Synteny visualization showing discrete flanking gene matches and how regions of interest were determined. Shows that the American Alligator has the largest number of identified and preserved genes within a scaffold, though this is likely aided by the fact that rAllMis1 is a chromosome-level assembly, rather than scaffold-level.

Discussion

- Among the species investigated, none had confirmed hits for PRDM9 presence
  - Aligns with implications from literature
- If both birds & crocodilians lack PRDM9, it could imply loss stemming from common ancestor

Table 1: Numerical results from pipeline.

Species	Flanking Genes (of 20)	Synteny (ρ)	Region of Interest Len. (kbp)	PRDM9 Found?
amer_gat	19	-0.46	9.63	No
chin_gat	9	0.98	12.85	No
aus_croc	7	0.63	43.9	No
gharial	3	1.00	398.08	Inconclusive

- **Limitations:**
  - Gharial "results" cannot be corroborated; did not run properly under current pipeline structure (may require manual analysis & oversight)
  - Synteny was determined through non-robust means; weak to paralogs



Future work

- Use more specialized synteny software (e.g., SYNY)
- Emphasize available chromosome-level assemblies
  - Would make results more interpretable & easier to locate
- Consider self-made annotations
  - E.g., Helixer, Tiberius, eggNOG-mapper
- Transfer current and future work to GitHub for version control
  - **Add in more genomes/species**



Citations

Data was sourced through NCBI Datasets CLI

Altschul, S. (1997). Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Research, 25(17), 3389–3402. https://doi.org/10.1093/nar/25.17.3389

Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., & Madden, T. L. (2009). BLAST+: architecture and applications. BMC Bioinformatics, 10(1). https://doi.org/10.1186/1471-2105-10-421

Lars Gabriel, Felix Becker, Katharina J Hoff, Mario Stanke, Tiberius: end-to-end deep learning with an HMM for gene prediction, Bioinformatics, Volume 40, Issue 12, December 2024, btae685, https://doi.org/10.1093/bioinformatics/btae685

GitHub Repository